

Trainer Debrief — Lab 17

Hypothesis Testing — In Depth, With Test Selection

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Test your capstone's candidate causes with correctly selected hypothesis tests (A3, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-17-hypothesis-testing.xlsx

TwoSample: $t \sim -3.24$, $p \sim 0.002$ — REJECT H_0 . The new fixture raises mean burst pressure by ~ 14.3 kPa. Check equal variances first (F-test $p > 0.05$, so pooled t is fine). Paired: mean difference $\sim +8.9$ kPa, $t \sim 8.0$, $p < 0.001$ — REJECT H_0 . Re-training works. A 2-sample t here would be WRONG: the observations are paired by operator. TwoProportion: Suzhou 8.8% vs Singapore 3.1%, $z \sim 7.1$, $p < 0.001$ — REJECT H_0 . Confirms the Lab 15 stratification finding statistically.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every test has a documented selection justification and assumption check, results report effect size alongside the p -value, and you maintain both a confirmed-cause and a rejected-cause log.

Where this sits in the capstone

A full re-derivation of hypothesis testing, including the test selection logic that Green Belts most often get wrong. You must be able to choose the right test, defend the choice, and explain the result in business language. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R ($\sim 40\%$ study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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